



Problem solving with integers and decimals

Task A: Integers, negatives and decimals in the real world



- 1) Aisha has £13.45 credit on her school account. She spends £2.67 then spends £7.89. Aisha then puts £4.21 back into her account.

How much does Aisha have in her school account now?

- 2) Jun has a £20 note. He buys 4 green pens, 6 pencils and 2 orange highlighters.

The shopkeeper gave Jun £6.16 in change.

Is this correct? Show your working out.



£1.56



76p



£1.27

- 3) There are two places in Antarctica scientists want to investigate. They look at the temperature in the morning over 5 days.



Location 1	-24.3°C	-21.6°C	-20.9°C	-24.15°C	-24.8°C
Location 2	-22°C	-21°C	-26°C	-24°C	-24.8°C

Using the mean average, which is the coldest location?

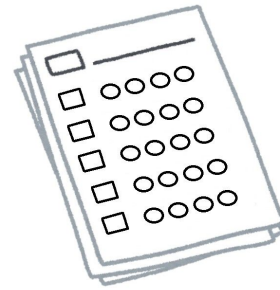
4) Andeep is taking part in a local Maths competition.

There are 10 questions.

Each correct answer gets 8 points.

Each incorrect answer gets (-6) points.

Each question not answered gets (-2) points.



a) Andeep answers 4 correct, 3 incorrect and misses 2 questions. How many points does he have?

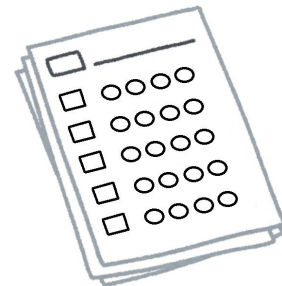
4b) In a different competition, Lucas gets 36 points.

There are 8 questions.

Each correct answer gets 12 points.

Each incorrect answer gets (-4) points.

Each question not answered gets (-2) points.



Give some examples of how Lucas performed in the competition.

Task B: Measurements and accuracy

1) Here is an example of two car rental companies.



Rent-a-Red-Car

1 km = £0.46



Rent-a-Blue-Car

2500 m = £1.10

An Oak teacher knows he will need to rent a car for a 70 km journey. Work out the difference in prices between the two companies for a 70 km car hire.

2) Sofia helps her neighbour feed her cats using a feeding chart.

Mass to the nearest 10 g	Quantity of dry food per day
 1 - 2.99 kg	70 g
 3 kg - 4.99 kg	90 g
 5 - 5.99 kg	100 g
 more than 6 kg	115 g



1.5 kg of dry food
£9.82

Her neighbour has three cats weighing 4.5 kg, 5.1 kg and 6.2 kg.

How much will it cost to feed all three cats for 2 weeks?